ASSIGNMENT-2

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40/50

① Linear Regression:-

Given $x = 1, 2, 3, 4, 5$

$y = 2.1, 3.9, 6.2, 7.8, 10.1$

ⓐ Find the regression equation:

Formula: $y = w_0 + w_1 x$

using the normal equations:

$$w_1 = \frac{S_{xy}}{S_{xx}} = \frac{19.9}{10} = 1.99$$

$$w_0 = \bar{y} - w_1 \bar{x} = 6.02 - 1.99(3) = 0.05$$

Regression equation:

$$y = 0.05 + 1.99x$$

ⓑ Predict at $x = 6$

$$y = 0.05 + 1.99(6) = 11.99$$

ⓒ Performance measures:-

- RSS = 0.107
- MSE = 0.0214
- $R^2 = 0.9973$

ⓓ Reliability:-

Prediction is reliable because R^2 is close to 1. however is outside the training range, so it is an extrapolation.

② Ridge Regression:-

Given

$S_{xx} = 10, S_{xy} = 19.9, \lambda = 5$

Ridge slope formula:

$$w = \frac{S_{xy}}{S_{xx} + \lambda}$$

$$w = \frac{19.9}{10+5} = 1.3267$$

least-squares slope = 1.99.

Percentage shrinkage:

$$\frac{1.99 - 1.3267}{1.99} \times 100 = 33.33\%$$

Ridge regression reduces the slope by 33.33%.

Increasing λ increases bias and reduces variance.

③ Naive Bayes classification:

(a) Calculate Prior Probabilities:

Total Spam: 5, Ham: 5.

P(Spam) = 5/10 = 0.5

P(Ham) = 5/10 = 0.5

Probability	Spam	Ham
Offer=Y	4/5	1/5
Offer=N	1/5	4/5
Link=Y	4/5	1/5
Link=N	1/5	4/5

(b) Offer=Y, Link=Y

$$P(Spam|X) = \frac{0.5(0.8)(0.8)}{0.5(0.8)(0.8) + 0.5(0.2)(0.2)}$$

$$\approx 0.9412 = 94.12\%$$

Classification: Spam.

(c) Laplace smoothing ($\alpha = 1$)

P(Offer=Y | Spam) = 5/7

P(Link=Y | Spam) = 5/7

Posterior:

$$P(Spam|X) = \frac{25}{29}$$

86.21%.

Classification: Spam.

(d) Offer=N, Link=N.

With smoothing

$$P(Spam|X) = \frac{4}{29} = 13.79$$

Classification: Ham.

Effect: Independence simplifies calculation. Smoothing prevents probabilities and improves predic

gistic Regression:-

Given:

$w_1 = 0, w_2 = 0, b = 0$

learning rate = 0.5

intial Probability = 0.5

$P = \sigma(0) = 0.5$

Gradient

$$\frac{\partial L}{\partial w_1} = 0.25$$

$$\frac{\partial L}{\partial w_2} = -0.25$$

$$\frac{\partial L}{\partial b} = 0$$

Updated weights:

$w_1 = 0 - 0.5(0.25) = -0.125$

$w_2 = 0 - 0.5(-0.25) = 0.125$

$b = 0$

Predicted Probabilites after update:-

sample 1 = 0.5312

sample 2 = 0.4688

Loss:- Before update = 0.6931

After update = 0.6326.

Result:- Loss decreases, showing that gradient descent improves the model.

5) Decision Tree - Entropy and information Gain:-

Given

Total samples = 10

yes = 6, No = 4.

(i) Entropy:-

$$H = -0.6 \log_2 0.6 - 0.4 \log_2 0.4$$

$$H = 0.971$$

(ii) Information Gain:-

child entropy:

- $A = a_1 : 0$
- $A = a_2 : 0.9183$

$IG = 0.971 - (0.4(0) + 0.6(0.9183))$

$IG = 0.420$

(iii) Gini impurity decrease

Parent Gini = 0.48

child Gini = 0.2667

Decrease = $0.48 - 0.2667$

0.2133

Conclusion:- The split is worthwhile because it reduces impurity.

(6) Cost Complexity Pruning

Given

- Subtree errors = 3
- leaves = 4
- Single leaf errors = 8

$R_\alpha(T) = R(T) + \alpha |\text{leaves}|$

$3 + 4\alpha = 8 + \alpha$

$3\alpha = 5$

$\alpha = 1.667$

3

For $a = 0.01$ and $a = 0.02$:

Both values are smaller than 1.667

Do Not Prune the subtree because it has lower cost.

Backpropagation:-

Given

$x_1 = 1, x_2 = 2$

$w_1 = 0.4, w_2 = 0.3, V = 0.5$

(i) forward Propagation:

$h = \sigma(0.4(1) + 0.3(2))$

$\boxed{h = 0.7311}$

$\hat{y} = \sigma(0.5 \times 0.7311)$

$\boxed{\hat{y} = 0.5904}$

(ii) Error:-

$E = \frac{1}{2}(1 - 0.5904)^2$

$\boxed{E = 0.0839}$

(iii) Back Propagation:-

Error terms:

$\delta_o = 0.09906$

$\delta_h = 0.009738$

Gradients:

$\partial E / \partial V = -0.07242$

$\partial E / \partial w_1 = -0.009738$

$\partial E / \partial w_2 = -0.019476$

(iv) Updated weights:-

$V = 0.5362$

$w_1 = 0.4049$

$w_2 = 0.3097$

New loss:

$\boxed{E = 0.0811}$

Conclusion:-

Backpropagation reduces the error.

(8) Support Vector Machine:-

$w = (0.5, 0.5), b = -2$

(a) Support Vectors:-

Decision function:

$f(x) = 0.5x_1 + 0.5x_2 - 2$

for (3,3)

$f(x) = 1$

for (1,1)

$f(x) = -1$

Both lie on margin hyperplanes.

Margin width.

$$\frac{2}{\|w\|} = \frac{2}{\sqrt{0.5}}$$

2.828

⑤ Classification:-

Point	output	class
(4,2)	1	Positive
(1,2)	-0.5	Negative

Distances:

* (4,2) = 1.414
* (1,2) = 0.707

⑥ Hinge loss:-

$$L = \max(0, 1 - y f(x))$$

* Point (1,2) : Loss = 0.5
* Point (2,2) : Loss = 1

Both violate the margin.

⑨ Ensemble learning

Given

individual classifier accuracy = 0.70

for 3 classifiers:-

$$P = 3(0.7)^2(0.3) + (0.7)^3$$

78.4%.

for 5 classifiers:-

$$P = \sum_{k=3}^{5} \binom{5}{k} (0.7)^k (0.3)^{5-k}$$

83.69%.

Conclusion:-

* Adding classifiers increases accuracy.
* strongly correlated errors reduce the benefit of majo

Voting.

Boost:-

Given

Total samples = 5.

Initial weight = 1/5 = 0.2

Misclassified samples = 2.

(i) weighted error

$$\epsilon = 2(0.2) = 0.4.$$

(ii) learned weight

$$\alpha = \frac{1}{2} \ln \frac{1-0.4}{0.4}$$

$$\boxed{\alpha = 0.2027}$$

(iii) updated weights:-

Correct samples:

$$0.2 e^{-0.2027} = 0.1633$$

Wrong samples

$$0.2 e^{0.2027} = 0.2449$$

(iv) Normalised weights:-

correct samples (1,3,4) = 0.1667 each.

wrong samples (2,5) = 0.25 each

Conclusion:- AdaBoost gives more weight to misclassified samples so the next learner focuses on them.

(11) Discriminant functions:-

Given $\mu_1 = (1,2)$, $\mu_2 = (3,1)$

$$\Sigma = \begin{bmatrix} 2 & 0.5 \\ 0.5 & 1 \end{bmatrix}$$

(a) find discriminant direction:-

$$\omega = \Sigma^{-1}(\mu_2 - \mu_1)$$

$$\omega = (1.4286 \quad -1.7143)$$

for equal priors, decision threshold

0.2857

(b) classify $x = (2.5, 1)$

$w^T x = 1.4286(2.5) - 1.7143(1)$

$= 1.8571$

since 1.8571 is greater than 0.2857;

(c) unequal Priors:-

$P(w_1) = 0.8$, $P(w_2) = 0.2$

New threshold

$0.2857 + \ln(4) = 1.672$

since 1.8571 is greater than 1.672

class 2 remains unchanged.

(12) Model Evaluation and Cost Analysis:-

Given three Models A, B and C.

(a) Performance measures:-

$$\text{Accuracy} = \frac{TP+TN}{\text{Total}} \qquad F_1 = \frac{2PR}{P+R}$$

$$\text{Precision} = \frac{TP}{TP+FP}$$

$$\text{Recall} = \frac{TP}{TP+FN}$$

Measure	Model A	Model B	Model C
Accuracy	88%	80.5%	93%
Precision	43.75%	32.08%	66.67%
Recall	70%	85%	60%
F1-score	53.85%	46.58%	63.16%

(b) Total expected cost:-

$$\text{Cost} = (FN \times 50000) + (FP \times 5000)$$

Model A:

$30(50000) + 90(5000)$

$= ₹19,50,000$

Model B:

$15(50000) + 180(5000)$

$= ₹16,50,000$

del C:
40(50000)+30(5000)
= 2 21,50,000

(c) Best Model:-
Model B should be deployed because it has the lowest total cost of 216,50,000 and highest recall (85%).
Model C has the highest accuracy and F1-score, but its total cost is highest.

(d) Improvement Steps:-
1. use cost-sensitive learning to reduce missed defaulters
2. Tune the classification threshold to minimise total cost.
Validation safeguard:- use k-fold cross-validation to prevent overfitting.